

Bansilal Ramnath Agarwal Charitable Trust's
Vishwakarma Institute of Technology, Pune-37

(An autonomous Institute of Savitribai Phule Pune University)



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1] DDL :- Data Definition Language

Data definition language is a category of SQL commands used to define, modify & manage the structure of database objects. It deals with the schema of the database rather than the actual data stored in it. DDL commands are responsible for creating, altering & deleting tables, view, indexes & constraints. When a DDL command is executed, the changes are automatically committed to the database, meaning they are permanent & cannot be rolled back in most cases.

Commands :-

CREATE - creates new db objects like tables, indexes.

```
CREATE TABLE students (  
    student_id SERIAL PRIMARY KEY,  
    name TEXT NOT NULL  
);
```

ALTER - Modifies existing db object.

```
ALTER TABLE students ADD COLUMN age INT;
```

DROP - Permanently deletes db objects.

```
DROP TABLE students;
```

TRUNCATE - Removes all rows.

```
TRUNCATE TABLE students;
```


2] DML :- Data Manipulation Language

Data Manipulation Language is a set of SQL commands used to manage & manipulate the data stored in database tables. Unlike DDL, which focuses on structures, DML works with the actual data within the tables. These commands allow inserting new records, updating existing ones, deleting unnecessary records, & retrieving data for analysis or processing.

Commands :-

INSERT - adds new rows of data

INSERT INTO students (name, age) **VALUES** ('Abhi', 19);

UPDATE - Modifies data.

UPDATE students **SET** age = 20 **WHERE** student_id = 1;

DELETE - Removes rows.

DELETE FROM students **WHERE** student_id = 1;

SELECT - Retrieves data from database.

SELECT * **FROM** students;

Title:- *handwritten title of project 7 2021*
Create a database with appropriate constraints using DDL & populate / modify it with the help of DML

Description: In this assignment, we applied various SQL operations on two tables — Available_Equipments and Exercises — demonstrating both DML and DDL commands. For Available_Equipments, we created the table with a primary key, inserted ten records, updated and deleted specific entries, added and modified a column, dropped it, and renamed both a column and the table. For Exercises, which has a foreign key referencing Available_Equipments, we inserted ten linked exercise records, performed updates and deletions, added and modified a column, removed it, and renamed a column and the table. These steps covered key SQL functions for managing and modifying relational databases.

Screenshots/Output:

Available Equipments Table:

The screenshot displays a database management tool interface. At the top, a tab labeled 'Equipment Inventory Setup' is active. Below it, a SQL editor shows the following code:

```
1 CREATE TABLE Available_Equipments (  
2   equipment_id SERIAL PRIMARY KEY,  
3   equipment_name TEXT NOT NULL  
4 );  
5
```

Below the SQL editor, the table 'public.available equipments' is selected. The table structure is shown with columns 'equipment_id' (int4) and 'equipment_name' (text). A '+ Table Creation' button is visible. The table is currently empty, with a message 'This table is empty' and an 'Import data from CSV' button. The bottom of the interface shows pagination controls: 'Page 1 of 1', '100 rows', and '0 records'. There are also 'Refresh', 'Data', and 'Definition' buttons.

Equipment Inventory Addition

```
1 INSERT INTO Available_Equipments (equipment_name) VALUES
2 ('Barbell'),
3 ('Squat Rack'),
4 ('Cable Machine'),
5 ('Dumbbells'),
6 ('Resistance Bands'),
7 ('Kettlebell'),
8 ('Treadmill'),
9 ('Stationary Bike'),
10 ('Pull-up Bar'),
11 ('Medicine Ball');
12
```

Insert 10 Rows

public.available equipments

Filter

Sort

Insert

RLS disabled

Role postgres

Realtime off

API Docs

	equipment_id	equipment_name	
	1	Barbell	
	2	Squat Rack	
	3	Cable Machine	
	4	Resistance Bands	
	5	Kettlebell	
	6	Treadmill	
	7	Stationary Bike	
	8	Pull-up Bar	
	9	Medicine Ball	
	10	Dumbbells	

Page 1 of 1

100 rows

10 records

Refresh

Data

Definition

Untitled query

```
1 UPDATE Available_Equipments
2 SET equipment_name = 'Adjustable Dumbbells'
3 WHERE equipment_id = 4;
4
```

Update

public.available equipments

Filter

Sort

Insert

RLS disabled

Role postgres

Realtime off

API Docs

☐	equipment_id	equipment_name	
☐	1	Barbell	
☐	2	Squat Rack	
☐	3	Cable Machine	
☐	4	Adjustable Dumbbells	
☐	5	Kettlebell	
☐	6	Treadmill	
☐	7	Stationary Bike	
☐	8	Pull-up Bar	
☐	9	Medicine Ball	
☐	10	Dumbbells	

Page 1 of 1

100 rows

10 records

Refresh

Data

Definition

Update Equipment Name

Equipment Removal

1 DELETE FROM Available_Equipments

2 WHERE equipment_id = 10;

3

Delete

☐	equipment_id	equipment_name	
☐	1	Barbell	
☐	2	Squat Rack	
☐	3	Cable Machine	
☐	4	Adjustable Dumbbells	
☐	5	Kettlebell	
☐	6	Treadmill	
☐	7	Stationary Bike	
☐	8	Pull-up Bar	
☐	9	Medicine Ball	

Page 1 of 1

100 rows

9 records

Refresh

Data

Definition

Add Equipment Category Column

1 ALTER TABLE Available_Equipments

2 ADD COLUMN category TEXT;

3

Add a column

public.available equipments

FilterSortInsert

RLS disabledRole postgresRealttime offAPI Docs

	equipment_id	equipment_name	category	
	1	Barbell	NULL	
	2	Squat Rack	NULL	
	3	Cable Machine	NULL	
	4	Adjustable Dumbbells	NULL	
	5	Kettlebell	NULL	
	6	Treadmill	NULL	
	7	Stationary Bike	NULL	
	8	Pull-up Bar	NULL	
	9	Medicine Ball	NULL	

Page 1 of 1100 rows9 recordsRefreshDataDefinition

Modify Equipment Category Field Length

1 ALTER TABLE Available_Equipments

2 ALTER COLUMN category TYPE VARCHAR(50);

3

Modify column type

public.available equipments

Filter Sort Insert RLS disabled Role postgres Realtime off API Docs

	equipment_id int4	equipment_name text	category varchar	+
☐	1	Barbell	NULL	
☐	2	Squat Rack	NULL	
☐	3	Cable Machine	NULL	
☐	4	Adjustable Dumbbells	NULL	
☐	5	Kettlebell	NULL	
☐	6	Treadmill	NULL	
☐	7	Stationary Bike	NULL	
☐	8	Pull-up Bar	NULL	
☐	9	Medicine Ball	NULL	

Page 1 of 1 100 rows 9 records Refresh Data Definition

Modify Equipment Category Field Length

Untitled query

```

1 ALTER TABLE Available_Equipments
2 DROP COLUMN category;
3

```

Drop a column

public.available equipments

Filter Sort Insert RLS disabled Role postgres Realtime off API Docs

	equipment_id int4	equipment_name text	+
☐	1	Barbell	
☐	2	Squat Rack	
☐	3	Cable Machine	
☐	4	Adjustable Dumbbells	
☐	5	Kettlebell	
☐	6	Treadmill	
☐	7	Stationary Bike	
☐	8	Pull-up Bar	
☐	9	Medicine Ball	

Page 1 of 1 100 rows 9 records Refresh Data Definition

Rename Equipment Column

```
1 ALTER TABLE Available_Equipments
2 RENAME COLUMN equipment_name TO name;
3
```

Rename Column

public.available equipments

Filter Sort Insert

RLS disabled Role postgres Realtime off API Docs

	equipment_id	name	
☐	1	Barbell	
☐	2	Squat Rack	
☐	3	Cable Machine	
☐	4	Adjustable Dumbbells	
☐	5	Kettlebell	
☐	6	Treadmill	
☐	7	Stationary Bike	
☐	8	Pull-up Bar	
☐	9	Medicine Ball	

Page 1 of 1 100 rows 9 records Refresh Data Definition

Rename Equipment Table

```
1 ALTER TABLE Available_Equipments
2 RENAME TO Gym_Equipments;
3
```

Rename Table

public.gym equipments

Filter

Sort

Insert

RLS disabled

Role postgres

Realtime off

API Docs

☐	equipm...	name	text	
☐	1	Barbell		
☐	2	Squat Rack		
☐	3	Cable Machine		
☐	4	Adjustable Dumbbells		
☐	5	Kettlebell		
☐	6	Treadmill		
☐	7	Stationary Bike		
☐	8	Pull-up Bar		
☐	9	Medicine Ball		

Page 1 of 1

100 rows

9 records

Refresh

Data

Definition

Exercises Table:

Exercise Tracking Table

```
1 CREATE TABLE Exercises (  
2   exercise_id SERIAL PRIMARY KEY,  
3   exercise_name TEXT NOT NULL,  
4   muscle_group TEXT,  
5   equipment_id INT NOT NULL,  
6   FOREIGN KEY (equipment_id) REFERENCES Available_Equipments(equipment_id)  
7   ON DELETE CASCADE  
8 );  
9
```

Table Creation

exercises

Filter

Sort

Insert

RLS disabled

Role postgres

Realtime off

API Docs

exercise_id

int4

exercise_name

text

muscle_group

text

equipment_id

int4

This table is empty

Import data from CSV

or drag and drop a CSV file here

Page 1 of 1

100 rows

0 records

Refresh

Data

Definition

Exercise Data Population

```
1 INSERT INTO Exercises (exercise_name, muscle_group, equipment_id) VALUES  
2 ('Bench Press', 'Chest', 1),  
3 ('Squat', 'Legs', 2),  
4 ('Deadlift', 'Back', 1),  
5 ('Bicep Curl', 'Biceps', 4),  
6 ('Tricep Pushdown', 'Triceps', 3),  
7 ('Shoulder Press', 'Shoulders', 4),  
8 ('Lat Pulldown', 'Back', 3),  
9 ('Leg Press', 'Legs', 2),  
10 ('Chest Fly', 'Chest', 4),  
11 ('Plank with Bands', 'Core', 5);  
12
```

Insert 10 Rows

public.exercises

Filter

Sort

Insert

RLS disabled

Role postgres

Realtime off

API Docs

	exercise_id int4	exercise_name text	muscle_group text	equipment_id int4	
☐	1	Bench Press	Chest	1	→
☐	2	Squat	Legs	2	→
☐	3	Deadlift	Back	1	→
☐	4	Bicep Curl	Biceps	4	→
☐	5	Tricep Pushdown	Triceps	3	→
☐	6	Shoulder Press	Shoulders	4	→
☐	7	Lat Pulldown	Back	3	→
☐	8	Leg Press	Legs	2	→
☐	9	Chest Fly	Chest	4	→
☐	10	Plank with Bands	Core	5	→

Page 1 of 1

100 rows 10 records

Refresh

Data

Definition

Update Exercise Muscle Group

```
1 UPDATE Exercises
2 SET muscle_group = 'Upper Chest'
3 WHERE exercise_id = 1;
4
```

Update

public.exercises

Filter

Sort

Insert

RLS disabled

Role postgres

Realtime off

API Docs

	exercise_id int4	exercise_name text	muscle_group text	equipment_id int4	
☐	1	Bench Press	Upper Chest	1	→
☐	2	Squat	Legs	2	→
☐	3	Deadlift	Back	1	→
☐	4	Bicep Curl	Biceps	4	→
☐	5	Tricep Pushdown	Triceps	3	→
☐	6	Shoulder Press	Shoulders	4	→
☐	7	Lat Pulldown	Back	3	→
☒	8	Leg Press	Legs	2	→
☐	9	Chest Fly	Chest	4	→
☐	10	Plank with Bands	Core	5	→

Page 1 of 1

100 rows 10 records

Refresh

Data

Definition

Remove Exercise #10

1

DELETE FROM Exercises

2

WHERE exercise_id = 10;

3

Delete

public.exercises

Filter

Sort

Insert

RLS disabled

Role postgres

Realtime off

API Docs

	exercise_id	exercise_name	muscle_group	equipment_id	
	1	Bench Press	Upper Chest	1	
	2	Squat	Legs	2	
	3	Deadlift	Back	1	
	4	Bicep Curl	Biceps	4	
	5	Tricep Pushdown	Triceps	3	
	6	Shoulder Press	Shoulders	4	
	7	Lat Pulldown	Back	3	
	8	Leg Press	Legs	2	
	9	Chest Fly	Chest	4	

Page 1 of 1

100 rows

9 records

Refresh

Data

Definition

Add Exercise Difficulty Level

1

ALTER TABLE Exercises

2

ADD COLUMN difficulty_level TEXT;

3

Add a column

public.exercises						
	exercise_id	exercise_name	muscle_group	equipment_id	difficulty_level	
	1	Bench Press	Upper Chest	1	NULL	
	2	Squat	Legs	2	NULL	
	3	Deadlift	Back	1	NULL	
	4	Bicep Curl	Biceps	4	NULL	
	5	Tricep Pushdown	Triceps	3	NULL	
	6	Shoulder Press	Shoulders	4	NULL	
	7	Lat Pulldown	Back	3	NULL	
	8	Leg Press	Legs	2	NULL	
	9	Chest Fly	Chest	4	NULL	

Modify Exercise Difficulty Format						
<pre> 1 ALTER TABLE Exercises 2 ALTER COLUMN difficulty_level TYPE VARCHAR(50); 3 </pre>						
Modify column type						
public.available equipments public.exercises						
	exercise_id	exercise_name	muscle_group	equipment_id	difficulty_level	
	1	Bench Press	Upper Chest	1	NULL	
	2	Squat	Legs	2	NULL	
	3	Deadlift	Back	1	NULL	
	4	Bicep Curl	Biceps	4	NULL	
	5	Tricep Pushdown	Triceps	3	NULL	
	6	Shoulder Press	Shoulders	4	NULL	
	7	Lat Pulldown	Back	3	NULL	
	8	Leg Press	Legs	2	NULL	
	9	Chest Fly	Chest	4	NULL	

Remove Difficulty Level

1

ALTER TABLE Exercises

2

DROP COLUMN difficulty_level;

3

Drop a column

public.exercises

FilterSortInsert

RLS disabledRole postgresRealtime offAPI Docs

	exercise_id int4	exercise_name text	muscle_group text	equipment_id int4	
	1	Bench Press	Upper Chest	1	→
	2	Squat	Legs	2	→
	3	Deadlift	Back	1	→
	4	Bicep Curl	Biceps	4	→
	5	Tricep Pushdown	Triceps	3	→
	6	Shoulder Press	Shoulders	4	→
	7	Lat Pulldown	Back	3	→
	8	Leg Press	Legs	2	→
	9	Chest Fly	Chest	4	→

Page 1 of 1100 rows9 recordsRefreshDataDefinition

Rename Exercise Column

1

ALTER TABLE Exercises

2

RENAME COLUMN exercise_name TO name;

3

Rename Column

public.exercises

Filter Sort Insert RLS disabled Role postgres Realtime off API Docs

exercise_id	muscle_group	equipment_id	name	
1	Upper Chest	1	Bench Press	
2	Legs	2	Squat	
3	Back	1	Deadlift	
4	Biceps	4	Bicep Curl	
5	Triceps	3	Tricep Pushdown	
6	Shoulders	4	Shoulder Press	
7	Back	3	Lat Pulldown	
8	Legs	2	Leg Press	
9	Chest	4	Chest Fly	

Page 1 of 1 100 rows 9 records Refresh Data Definition

Rename Exercises Table

```

1 ALTER TABLE Exercises
2 RENAME TO Gym_Exercises;
3

```

Rename Table

public.gym_exercises

Filter Sort Insert RLS disabled Role postgres Realtime off API Docs

exerci...	name	muscle_group	equipm...	
1	Bench Press	Upper Chest	1	
2	Squat	Legs	2	
3	Deadlift	Back	1	
4	Bicep Curl	Biceps	4	
5	Tricep Pushdown	Triceps	3	
6	Shoulder Press	Shoulders	4	
7	Lat Pulldown	Back	3	
8	Leg Press	Legs	2	
9	Chest Fly	Chest	4	

Page 1 of 1 100 rows 9 records Refresh Data Definition